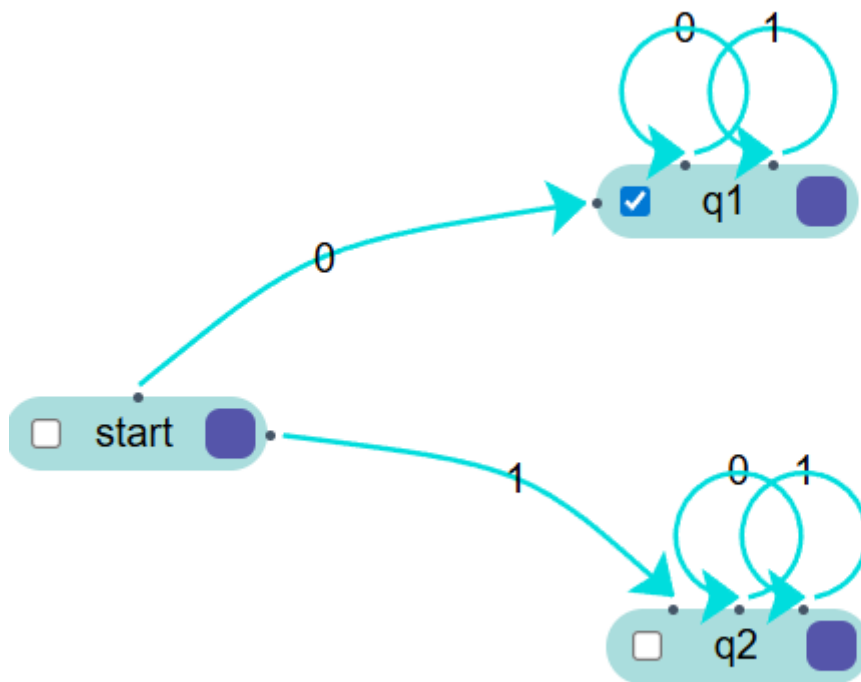




Test Results:

Accept: 01010 -- Pass
Accept: 01111 -- Pass
Accept: 00010 -- Pass
Accept: 01010 -- Pass
Accept: 00011 -- Pass
Accept: 10100 -- Fail
Accept: 11101 -- Fail
Accept: 10001 -- Fail
Accept: 11111 -- Fail
Accept: 10000 -- Fail
Reject: [Empty String] -- Pass

Tupla del AFD

$$A = (\Sigma, Q, \delta, q_0, F)$$

$$\Sigma = \{0,1\}$$

- $Q = \{q_0, q_1, q_2\}$
- q_0 = estado inicial
- $F = \{q_1\}$
- δ = función de transición

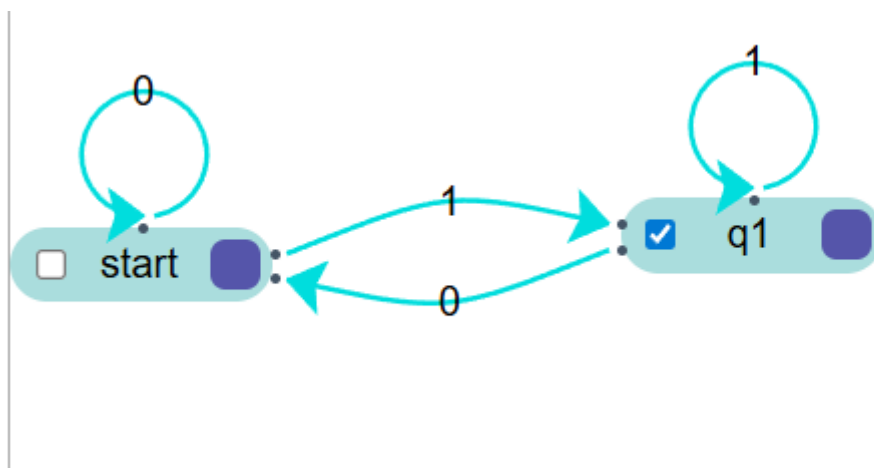
Estado 0 1

→ q0 q1 q2

* q1 q1 q1

q2 q2 q2

2-



Test Results:

Accept: 00001 -- Pass
Accept: 11111 -- Pass
Accept: 01011 -- Pass
Accept: 00011 -- Pass
Accept: 11101 -- Pass
Accept: 11010 -- Fail
Accept: 00000 -- Fail
Accept: 11110 -- Fail
Accept: 11000 -- Fail
Accept: 01000 -- Fail
Reject: [Empty String] -- Pass

Para este lenguaje:

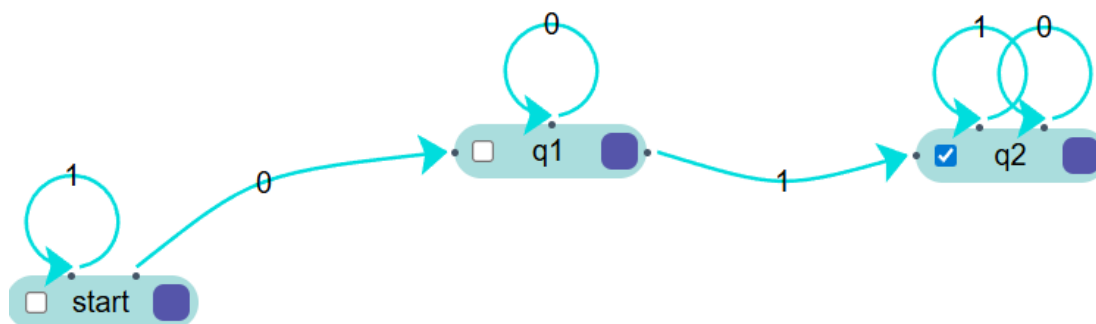
- $\Sigma = \{0,1\}$
- $Q = \{q_0, q_1\}$
- q_0 = estado inicial
- $F = \{q_1\}$

Estado 0 1

→ q0 q0 q1

* q1 q0 q1

3-



Test Results:

Accept: 10100 -- Pass
Accept: 01000 -- Pass
Accept: 11101 -- Pass
Accept: 10010 -- Pass
Accept: 11010 -- Pass
Accept: 10000 -- Fail
Accept: 111 -- Fail
Accept: 11000 -- Fail
Accept: 11111 -- Fail
Accept: 1 -- Fail
Reject: [Empty String] -- Pass

- $\Sigma = \{0,1\}$
- $Q = \{q_0, q_1, q_2\}$
- q_0 = estado inicial
- $F = \{q_2\}$

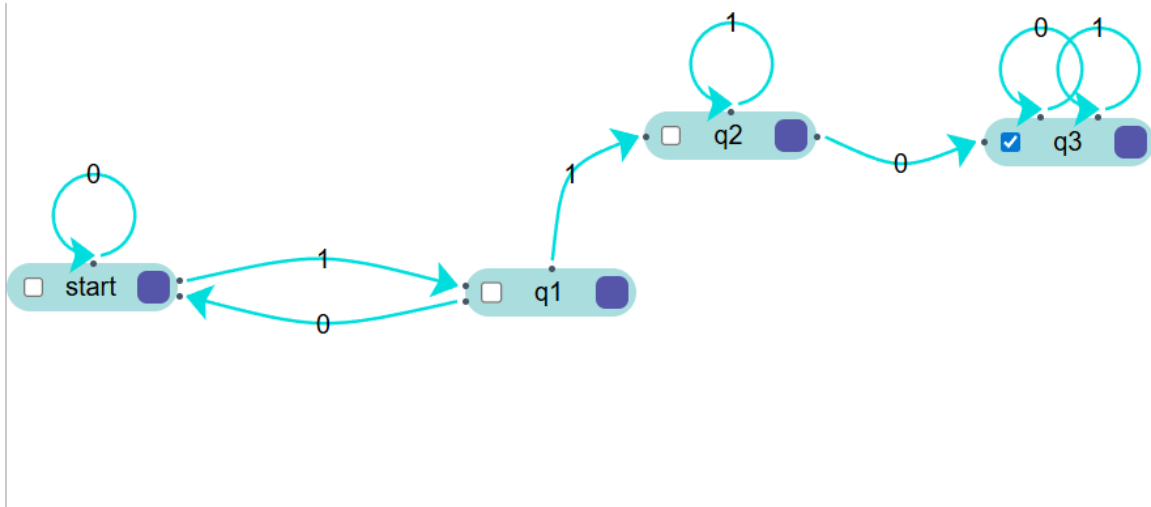
Estado 0 1

$\rightarrow q_0 \quad q_1 \quad q_0$

$q_1 \quad q_1 \quad q_2$

$* q_2 \quad q_2 \quad q_2$

4-



Test Results:

Accept: 01101 -- Pass

Accept: 11001 -- Pass

Accept: 00110 -- Pass

Accept: 11101 -- Pass

Accept: 10110 -- Pass

Accept: 01000 -- Fail

Accept: 1 -- Fail

Accept: 000 -- Fail

Accept: 10101 -- Fail

Accept: 00111 -- Fail

Reject: [Empty String] -- Pass

Estado 0 1

→ q0 q0 q1

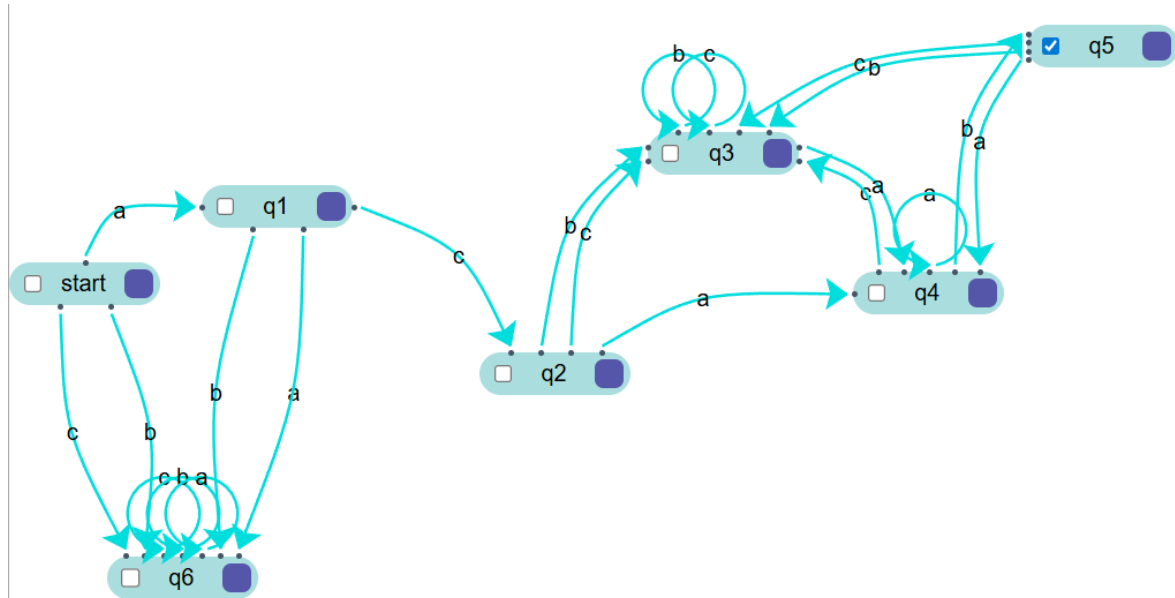
q1 q0 q2

q2 q3 q2

* q3 q3 q3

- $\Sigma = \{0,1\}$
- $Q = \{q0, q1, q2, q3\}$
- q0 = estado inicial
- $F = \{q3\}$

5-



Test Results:

Accept: acbab -- Pass
 Accept: acbbbab -- Pass
 Accept: acccbab -- Pass
 Accept: acbbbab -- Pass
 Accept: accbcab -- Pass
 Accept: abab -- Fail
 Accept: acaba -- Fail
 Accept: bacba -- Fail
 Accept: bacba -- Fail
 Accept: cabacab -- Fail
 Reject: [Empty String] -- Pass

Estado a b c

→ q0 q1 qt qt

q1 qt qt q2

q2 q4 q3 q3

q3 q4 q3 q3

q4 q4 q5 q3

* q5 q4 q3 q3

qt qt qt qt

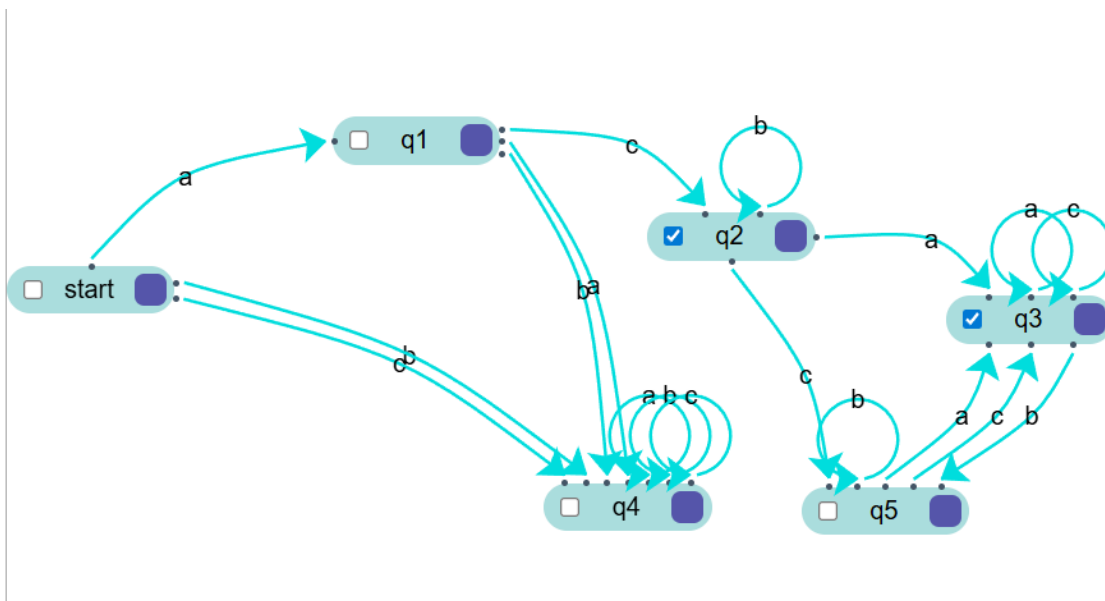
$\Sigma = \{a, b, c\}$

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_t\}$

q_0 = estado inicial

$F = \{q_5\}$

6-



Test Results:

Accept: ac -- Pass

Accept: acb -- Pass

Accept: aca -- Pass

Accept: acaa -- Pass

Accept: acba -- Pass

Accept: abbc -- Fail

Accept: aabc -- Fail

Accept: cbac -- Fail

Accept: acab -- Fail

Accept: accb -- Fail

Accept: [Empty String] -- Fail

Tupla del AFD

$$A = (\Sigma, Q, \delta, q_0, F)$$

$\Sigma = \{a, b, c\}$

$Q = \{\text{start}, q_1, q_2, q_3, q_4, q_5\}$

$q_0 = \text{start}$

$F = \{q_2, q_3\}$

Estado a b c

→ start q1 q4 q4

q1 q4 q4 q2

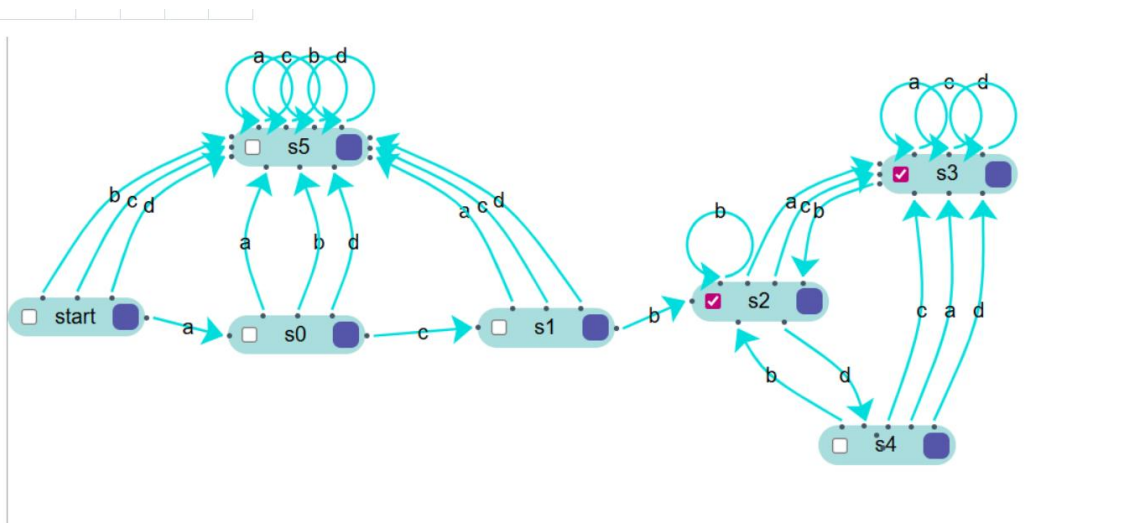
* q2 q3 q2 q5

* q3 q3 q5 q3

q4 q4 q4 q4

q5 q3 q5 q3

7-



Test Results:

Accept: acbd -- Fail
Accept: acbacbdba -- Pass
Accept: abcd -- Fail
Accept: acabddd -- Fail
Accept: acbbad -- Pass
Accept: adbacb -- Fail
Accept: acbbbbbd -- Fail
Accept: acb -- Pass
Accept: acbaddb -- Pass
Accept: acbdbbdb -- Pass
Reject: [Empty String] -- Pass

$$AFD=(\Sigma,Q,\delta,q_0,F)$$

Donde:

$$\Sigma = \{a,b,c,d\}$$

$$Q = \{\text{start}, s_0, s_1, s_2, s_3, s_4, s_5\}$$

$$q_0 = \text{start}$$

$$F = \{s_2, s_3\}$$

Función de transición δ :

$$\delta(\text{start},a) = s_0$$

$$\delta(\text{start},b) = s_5$$

$$\delta(\text{start},c) = s_5$$

$$\delta(\text{start},d) = s_5$$

$$\delta(s_0,a) = s_5$$

$$\delta(s_0,b) = s_5$$

$$\delta(s_0,c) = s_1$$

$$\delta(s_0,d) = s_5$$

$$\delta(s_1,a) = s_5$$

$$\delta(s_1,b) = s_2$$

$$\delta(s_1,c) = s_5$$

$$\delta(s_1,d) = s_5$$

$$\delta(s_2,a) = s_3$$

$$\delta(s_2,b) = s_2$$

$$\delta(s_2,c) = s_3$$

$$\delta(s_2,d) = s_4$$

$$\delta(s_3,a) = s_3$$

$$\delta(s_3,b) = s_2$$

$$\delta(s_3,c) = s_3$$

$$\delta(s_3,d) = s_3$$

$$\delta(s_4,a) = s_3$$

$$\delta(s_4,b) = s_2$$

$$\delta(s_4,c) = s_3$$

$$\delta(s_4,d) = s_3$$

$$\delta(s_5,a) = s_5$$

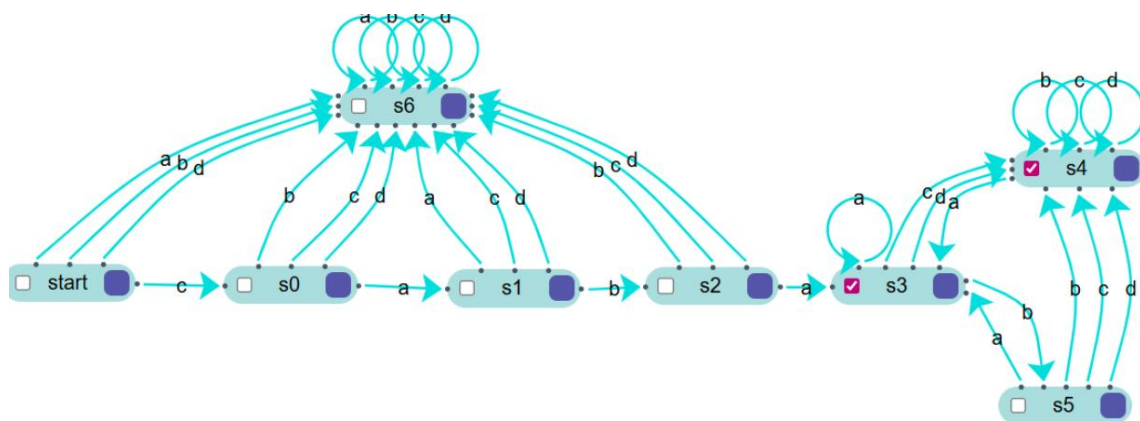
$$\delta(s_5,b) = s_5$$

$$\delta(s_5,c) = s_5$$

$$\delta(s_5,d) = s_5$$

Estado	a	b	c	d
→ start	s0	s5	s5	s5
s0	s5	s5	s1	s5
s1	s5	s2	s5	s5
* s2	s3	s2	s3	s4
* s3	s3	s2	s3	s3
s4	s3	s2	s3	s3
s5	s5	s5	s5	s5

8-



Test Results:

Accept: cabaab -- Fail
 Accept: caba -- Pass
 Accept: cabab -- Fail
 Accept: cababbcda -- Pass
 Accept: cababbbbab -- Fail
 Accept: bacabaa -- Fail
 Accept: cabaabdcca -- Pass
 Accept: cabaddaacc -- Pass
 Accept: ab -- Fail
 Accept: cababaca -- Pass
 Reject: [Empty String] -- Pass

Estado	a	b	c	d
→ start	s6	s6	s0	s6
s0	s1	s6	s6	s6
s1	s6	s2	s6	s6
s2	s3	s6	s6	s6
* s3	s3	s5	s4	s4
* s4	s3	s4	s4	s4
s5	s3	s4	s4	s4
s6	s6	s6	s6	s6

$$AFD=(\Sigma,Q,\delta,q_0,F)$$

Donde:

$$\Sigma = \{a,b,c,d\}$$

$$Q = \{\text{start}, s0, s1, s2, s3, s4, s5, s6\}$$

$$q_0 = \text{start}$$

$$F = \{s3, s4\}$$

Función de transición δ :

$$\delta(\text{start},a) = s6$$

$$\delta(\text{start},b) = s6$$

$$\delta(\text{start},c) = s0$$

$$\delta(\text{start},d) = s6$$

$$\delta(s0,a) = s1$$

$$\delta(s0,b) = s6$$

$$\delta(s0,c) = s6$$

$$\delta(s0,d) = s6$$

$$\delta(s1,a) = s6$$

$$\delta(s1,b) = s2$$

$$\delta(s1,c) = s6$$

$$\delta(s1,d) = s6$$

$$\delta(s2,a) = s3$$

$$\delta(s2,b) = s6$$

$$\delta(s2,c) = s6$$

$$\delta(s2,d) = s6$$

$$\delta(s3,a) = s3$$

$$\delta(s3,b) = s5$$

$$\delta(s3,c) = s4$$

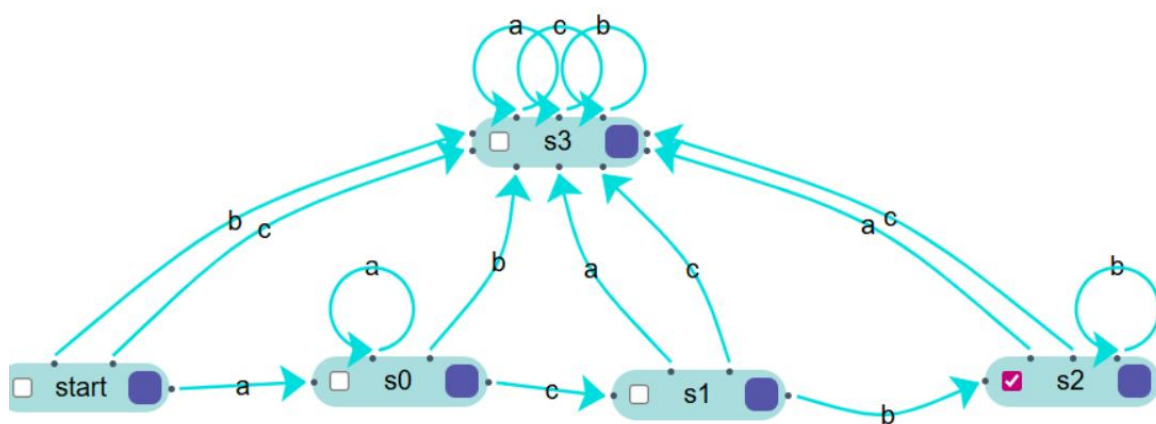
$$\delta(s3,d) = s4$$

$$\delta(s4,a) = s3$$

$$\delta(s4,b) = s4$$

$\delta(s4,c) = s4$
 $\delta(s4,d) = s4$
 $\delta(s5,a) = s3$
 $\delta(s5,b) = s4$
 $\delta(s5,c) = s4$
 $\delta(s5,d) = s4$
 $\delta(s6,a) = s6$
 $\delta(s6,b) = s6$
 $\delta(s6,c) = s6$
 $\delta(s6,d) = s6$

8-



Test Results:

Accept: aaaaaacbbbbbb -- Pass

Accept: acb -- Pass

Accept: ab -- Fail

Accept: acccb -- Fail

Accept: aaaacccbbb -- Fail

Accept: aacb -- Pass

Accept: acbbbbb -- Pass

Accept: bac -- Fail

Accept: acbbbac -- Fail

Accept: aaacbb -- Pass

Reject: [Empty String] -- Pass

Estado	a	b	c
→ start	s0	s3	s3
s0	s0	s3	s1
s1	s3	s2	s3
* s2	s3	s2	s3
s3	s3	s3	s3

$AFD=(\Sigma,Q,\delta,q_0,F)$

Donde:

$\Sigma = \{a,b,c\}$

$Q = \{\text{start}, s0, s1, s2, s3\}$

$q_0 = \text{start}$

$F = \{s2\}$

Función de transición δ :

$\delta(\text{start},a) = s0$

$\delta(\text{start},b) = s3$

$\delta(\text{start},c) = s3$

$\delta(s0,a) = s0$

$\delta(s0,b) = s3$

$\delta(s0,c) = s1$

$\delta(s1,a) = s3$

$\delta(s1,b) = s2$

$\delta(s1,c) = s3$

$\delta(s2,a) = s3$

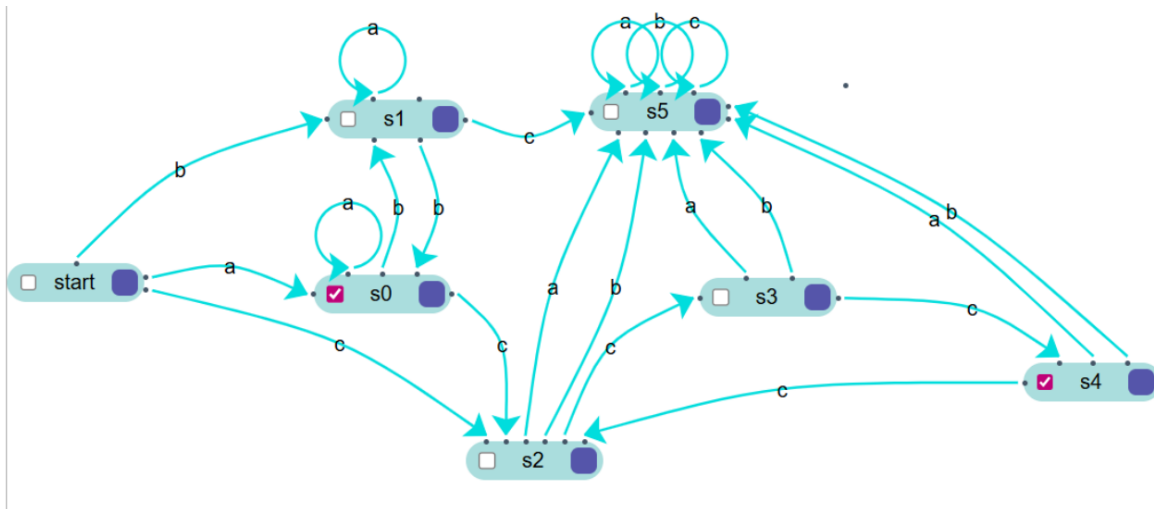
$\delta(s2,b) = s2$

$\delta(s2,c) = s3$

$\delta(s3,a) = s3$

$\delta(s3,b) = s3$

$\delta(s3,c) = s3$



Test Results:

Accept: abcc -- Fail
 Accept: ababbbccc -- Pass
 Accept: abccc -- Fail
 Accept: abb -- Pass
 Accept: caabbccc -- Fail
 Accept: bbbccc -- Pass
 Accept: bbbb -- Pass
 Accept: ccc -- Pass
 Accept: aaaacc -- Fail
 Accept: abbcccbab -- Fail
 Reject: [Empty String] -- Pass

Estado	a	b	c
→ start	s0	s1	s2
* s0	s0	s1	s2
s1	s1	s0	s5
s2	s5	s5	s3
s3	s5	s5	s4
* s4	s5	s5	s2
s5	s5	s5	s5

$AFD = (\Sigma, Q, \delta, q_0, F)$

Donde:

$\Sigma = \{a, b, c\}$

$Q = \{\text{start}, s0, s1, s2, s3, s4, s5\}$

$q_0 = \text{start}$

$F = \{s0, s4\}$

Función de transición δ :

$$\delta(\text{start}, a) = s_0$$

$$\delta(\text{start}, b) = s_1 \quad \delta(\text{start}, c) = s_2$$

$$\delta(s_0, a) = s_0$$

$$\delta(s_0, b) = s_1$$

$$\delta(s_0, c) = s_2$$

$$\delta(s_1, a) = s_1$$

$$\delta(s_1, b) = s_0$$

$$\delta(s_1, c) = s_5$$

$$\delta(s_2, a) = s_5$$

$$\delta(s_2, b) = s_5$$

$$\delta(s_2, c) = s_3$$

$$\delta(s_3, a) = s_5$$

$$\delta(s_3, b) = s_5$$

$$\delta(s_3, c) = s_4$$

$$\delta(s_4, a) = s_5$$

$$\delta(s_4, b) = s_5$$

$$\delta(s_4, c) = s_2$$

$$\delta(s_5, a) = s_5$$

$$\delta(s_5, b) = s_5$$

$$\delta(s_5, c) = s_5$$